

received from this satellite are also used by defence forces to conduct surgical strikes.

Types of sensors

There are many other sensors used in military applications. This article briefly covers some latest sensors used in military applications including active sensors, smart sensors, intelligent sensors, camera sensors, IR sensors, micro-electro-mechanical systems (MEMS), nano-sensors and wearable sensors.

Active sensors. The radar is an example of an active sensor. Active sensors rely on their own sources of radiation to illuminate objects so that the energy reflected and returned to the sensors can be measured. The radar operates in the microwave and radio wavelength regions of the electromagnetic spectrum.

One way to comprehend the active sensor is the ultra-wide band radar module such as the one shown in Fig. 2. It is based on Raspberry Pi 3 that supports various algorithms including distance detection, moving object detection, intrusion detection and vehicle detection.

The radar used in military environment uses the same aforementioned principle. High-tech radar systems are associated with digital signal processing, machine learning and are capable of extracting useful information from very high noise levels.

Smart and intelligent sensors. These sensors have rich features including user-friendly, easily accessible, flexible, multifunctional, good processing power, self-testing, self-diagnosis, self-compensation and more. High reliability is one of the key aspects of military specifications. Many manufacturers qualify for such specifications, having reliability to a degree that consumer markets generally do not demand, and the conditions that are unique to military applications.

Visual information plays an important part in intelligent decision making for humans as well as machines. Electronic visual intelligence depends on the image sensors in camera systems. For instance, short-wave IR (SWIR) cameras offer a new weapon in the



Fig. 2: Radar module (Credit: www.umin.co.kr)



Fig. 3: DIGIC 4 image processor (Credit: <https://en.wikipedia.org>)

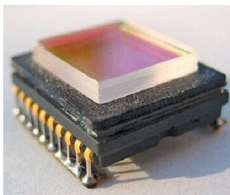


Fig. 4: Three-layer IR low-pass filter on CCD sensor (Credit: <https://en.wikipedia.org>)

war on terror and other combat operations.

Camera sensors. Key features of image sensors include face, motion and scene detection. Motion detection technology is used to detect the speed and direction of movement. It limits the effect of blurring of the subject, and scene detection technology determines the optimal settings including brightness, contrast and distance to the subject and colour of the entire image.

Viooa Imaging Technology has been producing smart camera sensors

for use in various military applications for unmanned platforms including air, ground and sea. These are designed for use in extreme indoor and outdoor harsh environments to get one or multiple points of interest from different directions.

Canon DIGIC 5 image sensor has amazing features that can achieve a capture rate of 10.3 frames per second at full resolution in high-speed burst HQ, full-HD 1080-pixel videos and intelligent image stabilisation. The company claims that this processor is six times faster than DIGIC 4 and efficiently manages the scene information and reduces noise by up to 75 per cent. This image sensor was co-designed with Texas Instruments (TI) and manufactured by TI through foundry partnership in Asia using 45nm node technology fanned out by TI. DIGIC 8 was introduced in February 2018.

A wide range of military cameras can be found on www.imperx.com/military-cameras/

IR sensors. These are used in image detection and IR jamming of a wide range of military equipment and vehicles including fighter aircraft, unmanned aerial vehicles (UAVs), helicopters, hexacopters, tanks, artillery, navy ships, submarines and missiles. Modern IR sensors can detect minute details at distances of over 100 metres. This technology enables aerial vehicles to fly in the darkest of night, while maintaining near-day visibility.

Many objects such as humans, vehicle engines and aircraft generate and emit heat, and so are visible in the IR wavelengths of light. A passive weapon guidance system or IR homing device uses IR light emission from a target to track and follow it.

Charge-couple device (CCD) and complementary metal oxide semiconductor (CMOS) sensors are used in digital cameras. Both sensors have relatively similar IR spectral response ranges, but CCD is more dominant because it produces superior images using fabrication technology.

MEMS sensors. These sensors could be deployed in airports, military camps, public buildings and other strategic locations. Miniaturised MEMS sensors have become popular for military applications. Many agen-